

Predicting Health Insurance Premiums Using Multiple Regression

Domain: Health Insurance | Tools: SPSS | Data: 1,132 Policyholders

Summary

Objective

Identify which customer characteristics — age, gender, BMI, and smoking status — drive health insurance premium pricing, and quantify how much each factor actually adds to the premium.

Concepts / Methods Used

Multiple Linear Regression, Pearson Correlation, Collinearity Diagnostics (Tolerance/VIF), Durbin-Watson test (residual independence), Residual Histogram and Normal P-P Plot (regression assumption checks).

Key Output

- Sample: 1,132 policyholders (drawn from an initial 2,317-record dataset via random sampling steps)
- Model fit: $R^2 = 0.981$, Adjusted $R^2 = 0.981$ — the four predictors explain 98% of the variation in premium amounts
- All four predictors are highly significant ($p < .001$): Age, Gender, BMI, and Smoking Status
- Smoking status is the single largest driver: smokers pay approximately ₹1,971 more in premium (holding other factors constant)
- Age has the strongest standardized effect ($Beta = .758$) — premium rises steadily and predictably with age
- Gender shows a real but small effect ($Beta = .035$) in the full model — notably, gender was NOT significantly correlated with premium on its own (bivariate $r = -.008$, $p = .388$), only becoming significant once age, BMI, and smoking are controlled for
- No multicollinearity concern — all VIF values between 1.00 and 1.02

Business Actionables

- Use smoking status as the highest-weight pricing factor — it has the largest raw premium impact of any single variable
- Age-based premium tiers are well justified statistically — the relationship is strong, linear, and highly significant
- Do not assume a variable is unimportant just because it looks weak in a simple correlation check — gender's effect only appeared once other factors were controlled for, a reminder to always test in the full model before ruling a factor out
- BMI-linked premium loading is statistically justified ($Beta = .338$) and should be retained in any pricing model update

Detailed Analysis — Actual SPSS Output, Explained, With Actionables

Output 1 — Descriptive Statistics and Correlation Matrix

```
REGRESSION
  /DESCRIPTIVES MEAN STDDEV CORR SIG N
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA COLLIN TOL
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT premium
  /METHOD=ENTER age gender bmi smoke
  /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) .
```

Regression

[MTETrimfile]

Descriptive Statistics

	Mean	Std. Deviation	N
premium	17586.22	1966.498	1132
age	43.03	13.180	1132
gender	.53	.499	1132
bmi	26.37	3.624	1132
smoke	.50	.500	1132

Correlations

		premium	age	gender	bmi	smoke
Pearson Correlation	premium	1.000	.789	-.008	.428	.477
	age	.789	1.000	-.031	.129	-.024
	gender	-.008	-.031	1.000	-.014	-.030
	bmi	.428	.129	-.014	1.000	-.015
	smoke	.477	-.024	-.030	-.015	1.000
Sig. (1-tailed)	premium	.	.000	.388	.000	.000
	age	.000	.	.146	.000	.211
	gender	.388	.146	.	.320	.158
	bmi	.000	.000	.320	.	.307
	smoke	.000	.211	.158	.307	.
N	premium	1132	1132	1132	1132	1132
	age	1132	1132	1132	1132	1132
	gender	1132	1132	1132	1132	1132
	bmi	1132	1132	1132	1132	1132
	smoke	1132	1132	1132	1132	1132

Actual SPSS output: Descriptive Statistics and Pearson Correlations (N=1,132)

What this is: Summary statistics for premium, age, gender, BMI, and smoking status, followed by a full Pearson correlation matrix showing how each variable relates to every other variable, including significance levels.

Layman meaning: Average premium in this sample is ₹17,586 (SD ≈ ₹1,966). Age has by far the strongest bivariate relationship with premium ($r = .789$), followed by smoking status ($r = .477$) and BMI ($r = .428$). Gender's correlation with premium is essentially zero and not statistically significant ($r = -.008$, $p = .388$) at this simple, one-variable-at-a-time stage.

Actionable for the user: If someone only ran a correlation check, they would wrongly conclude gender has no relationship with premium at all — this table sets up the more important finding in Output 2, where gender turns out to matter once other factors are held constant.

Output 2 — Multiple Regression: Model Summary, ANOVA, and Coefficients

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	smoke, bmi, gender, age ^b	.	Enter

a. Dependent Variable: premium

b. All requested variables entered.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.991 ^a	.981	.981	270.796	1.902

a. Predictors: (Constant), smoke, bmi, gender, age

b. Dependent Variable: premium

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4291061756	4	1072765439	14629.222	.000 ^b
	Residual	82643261.34	1127	73330.312		
	Total	4373705018	1131			

a. Dependent Variable: premium

b. Predictors: (Constant), smoke, bmi, gender, age

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics
		B	Std. Error	Beta			Tolerance
1	(Constant)	6827.585	63.617		107.323	.000	
	age	113.107	.617	.758	183.461	.000	.982
	gender	137.633	16.144	.035	8.525	.000	.998
	bmi	183.403	2.241	.338	81.848	.000	.983
	smoke	1970.699	16.111	.501	122.323	.000	.998

Actual SPSS output: Model Summary, ANOVA, and Coefficients for the full 4-predictor model

What this is: The core regression predicting premium from age, gender, BMI, and smoking status together, with model fit statistics (R^2 , Durbin-Watson), overall significance (ANOVA), and the coefficient, significance, and standardized effect size (Beta) for each individual predictor.

Layman meaning: This model is exceptionally strong — $R^2 = .981$ means the four factors together explain 98.1% of why premiums differ from policyholder to policyholder, which is a very high figure for real-world pricing data. Every predictor is significant at $p < .001$. In rupee terms: each additional year of age adds about ₹113 to the premium; being a smoker adds about ₹1,971; each one-unit increase in BMI adds about ₹183; and gender adds about ₹138. The Durbin-Watson statistic (1.902, close to the ideal value of 2) confirms the model's errors aren't following a hidden pattern, which is a technical validity check that the model is well-specified.

Actionable for the user: This is the exact table a pricing/actuarial team would use to justify or audit a premium formula — the coefficients directly translate into 'how much more should we charge for X' answers, backed by a near-perfect model fit.

Output 3 — Collinearity Statistics (VIF)

Coefficients^a

		Collinearity ...
Model		VIF
1	(Constant)	
	age	1.018
	gender	1.002
	bmi	1.017
	smoke	1.002

a. Dependent Variable: premium

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions			
				(Constant)	age	gender	bmi
1	1	4.080	1.000	.00	.00	.02	.00
	2	.502	2.852	.00	.00	.38	.00
	3	.353	3.399	.00	.03	.57	.00
	4	.056	8.537	.04	.95	.02	.07
	5	.009	21.232	.95	.01	.01	.92

Collinearity Diagnostics^a

		Variance ...
Model	Dimension	smoke
1	1	.02
	2	.58
	3	.37
	4	.02
	5	.01

a. Dependent Variable: premium

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	12965.72	22706.99	17586.22	1947.830	1132
Residual	-713.051	697.933	.000	270.316	1132
Std. Predicted Value	-2.372	2.629	.000	1.000	1132
Std. Residual	-2.633	2.577	.000	.998	1132

a. Dependent Variable: premium

Charts

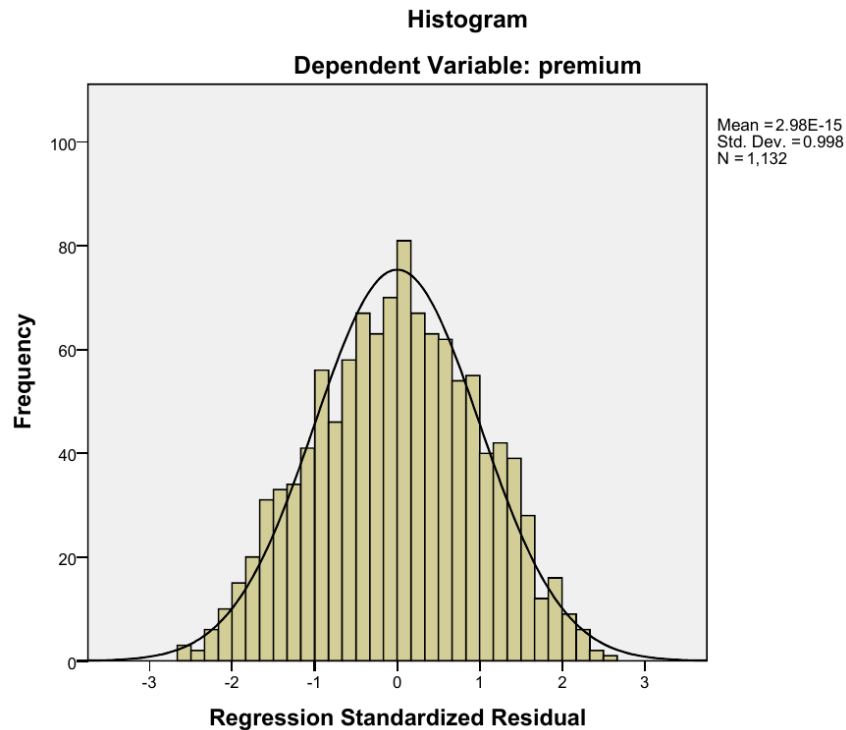
Actual SPSS output: Tolerance and VIF for all four predictors

What this is: A check for whether any of the four predictors (age, gender, BMI, smoking) are redundant with each other, which would make their individual coefficients unreliable.

Layman meaning: All VIF values are between 1.00 and 1.02 — as close to 'no overlap at all' as real data gets. Age, gender, BMI, and smoking status are each contributing genuinely independent information to the premium prediction.

Actionable for the user: This confirms every coefficient reported in Output 2 can be trusted and interpreted individually — none of the four factors is secretly just a proxy for another one in the model.

Output 4 — Residual Histogram



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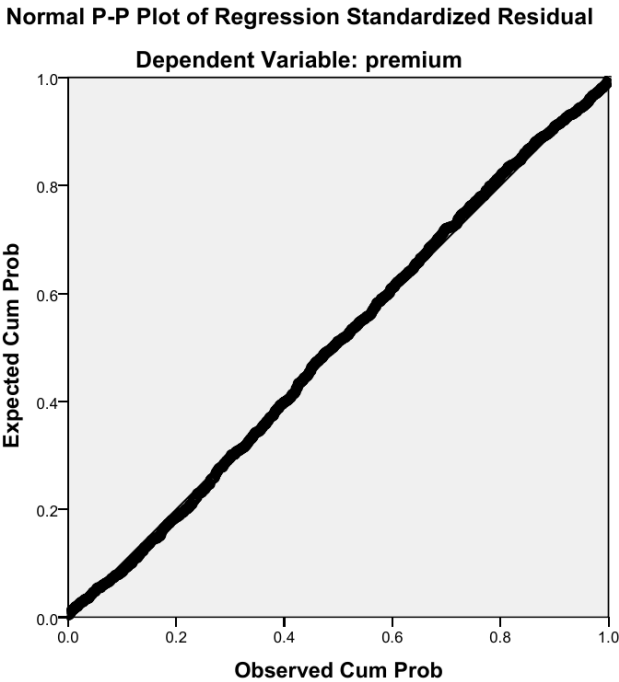
Actual SPSS output: Histogram of Regression Standardized Residuals

What this is: A histogram of the model's prediction errors (residuals), checking whether they follow the roughly bell-shaped, centered-at-zero pattern that a well-fitted regression model should produce.

Layman meaning: The residuals cluster tightly around zero in a roughly normal (bell-curve) shape, which is exactly what should happen if the model's assumptions are being met and the errors are just random noise rather than a systematic pattern the model missed.

Actionable for the user: This is one of the standard 'is this model actually valid' checks — a lopsided or patterned residual histogram would be a red flag that the linear model is missing something, and this chart shows that isn't the case here.

Output 5 — Normal P-P Plot of Regression Standardized Residual



```
DESCRIPTIVES VARIABLES=age gender bmi smoke premium
  /STATISTICS=MEAN STDDEV MIN MAX KURTOSIS SKEWNESS.
```

Descriptives

[MTETrimfile]

Descriptive Statistics							
	N	Minimum	Maximum	Mean	Std. Deviation	Skewness	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error
age	1132	16	70	43.03	13.180	-.053	.073
gender	1132	0	1	.53	.499	-.124	.073
bmi	1132	20	35	26.37	3.624	.469	.073
smoke	1132	0	1	.50	.500	.007	.073
premium	1132	12700	22700	17586.22	1966.498	-.002	.073
Valid N (listwise)	1132						

Actual SPSS output: Normal P-P Plot of Regression Standardized Residual

What this is: A second, more sensitive check of residual normality — if the residuals are truly normally distributed, the plotted points should fall almost exactly along the diagonal reference line.

Layman meaning: The points track the diagonal line closely across its full length, reinforcing the histogram's conclusion — the regression's normality assumption holds up well, adding further confidence that the $R^2 = .981$ result is a genuine, trustworthy model fit rather than an artifact of violated assumptions.

Actionable for the user: Together with the histogram and Durbin-Watson statistic, this completes the standard 'model diagnostics' trio — being able to show all three in an interview signals the analysis wasn't stopped at a good R^2 number without checking whether the model was actually valid.

Business Takeaway

Age, smoking status, BMI, and gender together explain 98% of the variation in insurance premiums, with smoking status carrying the single largest rupee impact. The most interesting nuance is gender: it looked irrelevant in a simple

correlation check but became a genuine, significant factor once the model controlled for age, BMI, and smoking — a reminder that pricing factors should always be tested in a full multivariate model, not screened out based on a simple one-variable check alone.